Seria Furie

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jk\omega_0 t}$$

$$|C_n| = \sqrt{Re^2[C_n] + Im^2[C_n]}$$

$$C_n = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jn\omega_0 t} dt$$

$$e^{-j2\pi k} = 1$$

$$e^{\mp j\pi} = -1$$

$$C_0 = \frac{1}{T} \int_{T} x(t) dt$$

(5.4) Të shënohet forma eksponenciale e seria furie për sinjalet e dhënë

a) $x(t) = \cos\omega_0 t$ → forma eksponenciale

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

e zbërthejm sinjalin e dhënë

$$x(t) = \frac{1}{2} e^{j\omega_0 t} + \frac{1}{2} e^{-j\omega_0 t}$$

$\downarrow$ $n=1$ $\quad$ $\downarrow$ $n=-1$

$$C_1 = \frac{1}{2} \qquad C_{-1} = \frac{1}{2}$$

b) $x(t) = \sin \omega_0 t$

$$x(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk\omega_0 t}$$

$$x(t) = \frac{1}{2j} e^{j\omega_0 t} - \frac{1}{2j} e^{-j\omega_0 t}$$

$n = 1$ $\qquad$ $n = -1$

$$C_1 = \frac{1}{2j} \qquad C_{-1} = -\frac{1}{2j}$$

Özel

$$C_1 = \frac{1}{2} e^{-j\pi/2} \qquad C_{-1} = \frac{1}{2} e^{j\pi/2}$$

c) $x(t) = \cos\left(2t + \frac{\pi}{4}\right)$

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

$$x(t) = \frac{1}{2} e^{j\left(2t + \frac{\pi}{4}\right)} + \frac{1}{2} e^{-j\left(2t + \frac{\pi}{4}\right)}$$

$$x(t) = \frac{1}{2} e^{j2t} \cdot e^{j\frac{\pi}{4}} + \frac{1}{2} e^{-j2t} e^{-j\frac{\pi}{4}}$$

$n = 1$ $\qquad$ $n = -1$

$$C_1 = \frac{1}{2} e^{j\pi/4} \qquad C_{-1} = \frac{1}{2} e^{-j\pi/4}$$

d) $x(t) = \cos 4t + \sin 6t$

$\omega_1 = 4 \qquad \omega_2 = 6$

$T_1 = \frac{2\pi}{\omega_1} = \frac{2\pi}{4} = \frac{\pi}{2}$; $T_2 = \frac{2\pi}{\omega_2} = \frac{2\pi}{6} = \frac{\pi}{3}$

$\frac{T_1}{T_2} = \frac{\frac{\pi}{2}}{\frac{\pi}{3}} = \frac{3}{2}$

$T_0 = 2T_1 = 3T_2$

$2 \cdot \frac{\pi}{2} = 3 \cdot \frac{\pi}{3}$

$\pi = \pi$

$\boxed{T_0 = \pi}$

$\omega = \frac{2\pi}{T_0} = \frac{2\pi}{\pi} = 2$

$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$

$= \sum_{n=-\infty}^{\infty} C_n e^{jn2t}$

$x(t) = \frac{1}{2} e^{j4t} + \frac{1}{2} e^{-j4t} + \frac{1}{2j} e^{j6t} - \frac{1}{2j} e^{-j6t}$

$\downarrow n=2 \qquad \downarrow n=-2 \qquad \downarrow n=3 \qquad \downarrow n=-3$

$C_2 = \frac{1}{2} \qquad C_{-2} = \frac{1}{2}$; $C_3 = \frac{1}{2j} \qquad C_{-3} = -\frac{1}{2j}$

e) $x(t) = \sin^2 t$

$$x(t) = \frac{1}{2} - \frac{1}{2}\cos 2t$$

↓

sinjal periodik me periode" të" gjendshme.

$$x(t) = \sum_{n=-\infty}^{\infty} c\, e^{jn2t}$$

$$x(t) = \frac{1}{2} - \frac{1}{2}\left(\frac{1}{2}e^{j2t} + \frac{1}{2}e^{-j2t}\right)$$

$$= \frac{1}{2} - \frac{1}{4}e^{j2t} - \frac{1}{4}e^{-j2t}$$

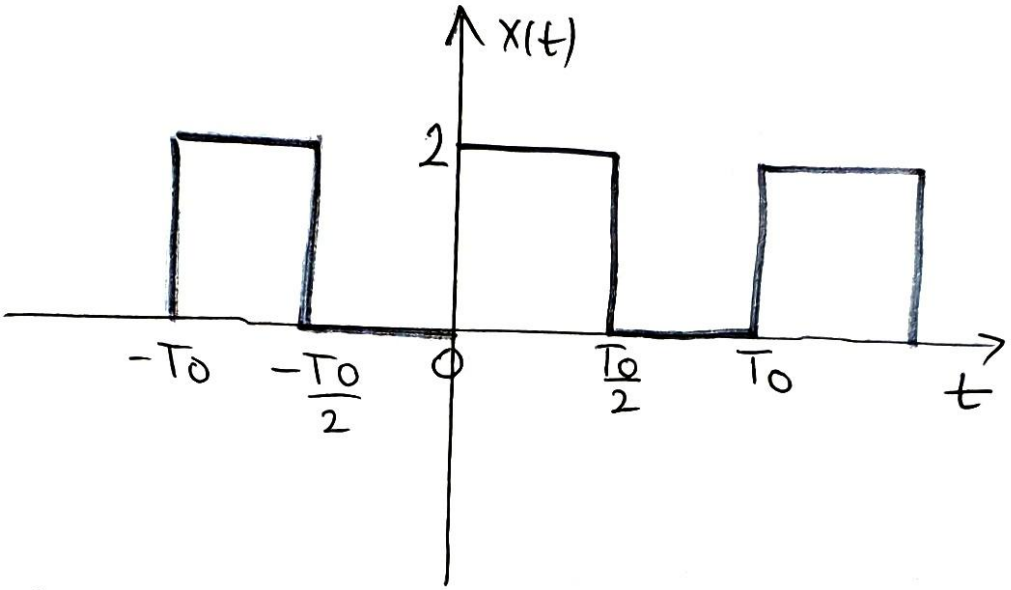
↓ $n=0$; ↓ $n=1$; ↓ $n=-1$

$$C_0 = \frac{1}{2}, \quad C_1 = -\frac{1}{4}, \quad C_{-1} = -\frac{1}{4}$$

5.5.

a

Të shkruhet forma eksponenciale për sinjalin x(t) të dhënë në figurë.



Zgjidhje

Periода e sinjalit është $T = T_o$; $\omega = \frac{2\pi}{T} = \frac{2\pi}{T_o}$

$$C_n = \frac{1}{T}\int_T x(t) e^{-jn\omega_o t}\, dt$$

Në mastin tonë kemi:

$$C_n = \frac{1}{T_o}\int_0^{T_o/2} 2\, e^{-jn\omega_o t}\, dt = \frac{2}{T_o}\int_0^{T_o/2} e^{-jn\omega_o t}\, dt$$

$$= \frac{2}{T_o}\left(-\frac{1}{jn\omega_o} e^{-jn\omega_o t}\Big|_0^{T_o/2}\right)$$

$$= \frac{2}{T_0}\left[\frac{1}{jn\omega_0}\left(e^{-jn\omega_0\frac{T_0}{2}} - e^{0}\right)\right]^{1} \qquad e^{-jn\pi}$$

zérendesgim e

$$= \frac{2}{T_0}\left(\frac{1 - e^{-jn\frac{\omega_0 T_0}{2}}}{jn\omega_0}\right)$$

T_0 → zérendesgim me $\frac{2\pi}{\omega}$

zérendesgim $(-1)^n$

$$= \frac{2}{\frac{2\pi}{\omega_0}}\left(\frac{1 - e^{-jn\pi}}{jn\omega_0}\right) = \frac{1}{j\pi n}\left(1 - (-1)^n\right)$$

$$= \boxed{\frac{(1-(-1)^n)}{j\pi n}}$$

Për

$n = 2k \rightarrow$ çift.

$$\boxed{C_{2k} = 0}$$

Për

$n = 2k+1$

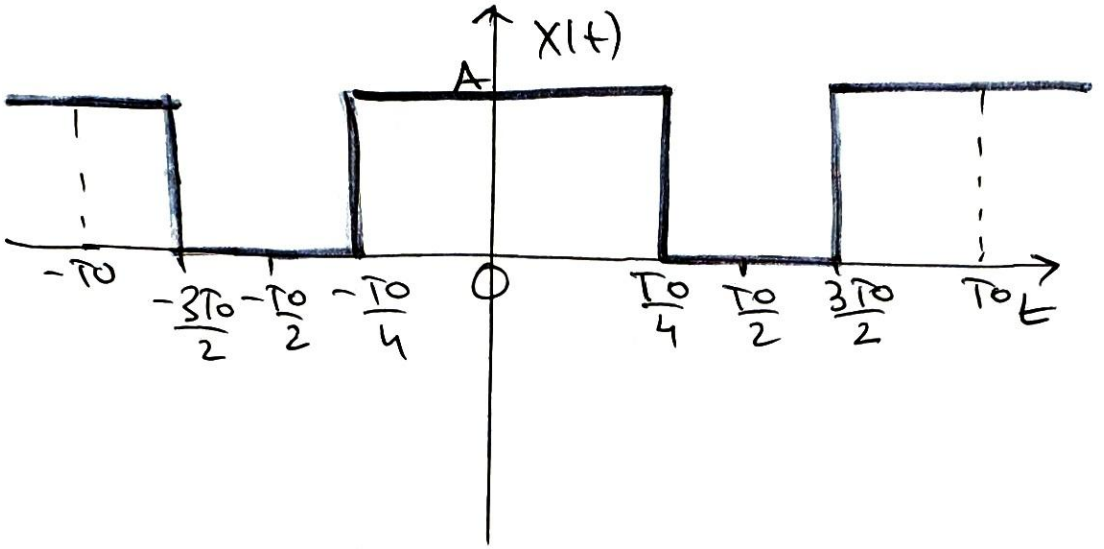
$$C_{2k+1} = \frac{2}{j\pi(2k+1)}$$

$$C_0 = \frac{1}{T_0}\int_0^{T_0/2} 2\,dt = \frac{2}{T_0}\cdot\frac{T_0}{2} = 1$$

$$x(t) = 1 + \sum_{k=-\infty}^{\infty} \frac{2}{j\pi(2k+1)} e^{j(2k+1)\omega_0 t}$$

5.6

Të shkruhet forma eksponenciale për sinjalin $x(t)$ të dhënë në figurë:



Zgjidhje

Perioda e sinjalit është $T = T_0$; $\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{T_0}$

Forma eksp: $x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$

$$C_n = \frac{1}{T} \int_T x(t) e^{-jn\omega_0 t} dt$$

Në rastin tonë:

$$C_n = \frac{1}{T_0} \int_{-\frac{T_0}{4}}^{\frac{T_0}{4}} A e^{-jn\omega_0 t} dt = \frac{A}{T_0} \left(\frac{-1}{jn\omega_0} e^{-jn\omega_0 t} \Big|_{-\frac{T_0}{4}}^{\frac{T_0}{4}} \right)$$

$$= \frac{A}{\frac{2\pi}{\omega_0}} \left(\frac{-1}{jn\omega_0} \left(e^{-jn\omega_0 \frac{T_0}{4}} - e^{jn\omega_0 \frac{T_0}{4}} \right) \right)$$

$-jn\omega_0 \frac{T_0}{4} \to -\frac{\pi}{2}$; $jn\omega_0 \frac{T_0}{4} \to \frac{\pi}{2}$

$$= \frac{A}{2\pi(jn)}\left(e^{jn\frac{\pi}{2}} - e^{-jn\frac{\pi}{2}}\right) = \frac{A}{\pi n}\left(\frac{e^{jn\frac{\pi}{2}} - e^{-jn\frac{\pi}{2}}}{2j}\right)$$

$$= \frac{A}{\pi n}\sin n\frac{\pi}{2}$$

Pën $n = 2k \rightarrow$ çift

$C_{2k} = 0$

Pën $n = 2k+1 \rightarrow$ tek

$$C_{2k+1} = \frac{A(-1)^k}{\pi(2k+1)}$$

$$C_0 = \frac{1}{T_0}\int_{-\frac{T_0}{4}}^{T_0/4} A\,dt = \frac{A}{2}$$

$$x(t) = \frac{A}{2} + \sum_{k=-\infty}^{\infty} \frac{A(-1)^k}{\pi(2k+1)} e^{j(2k+1)\omega_0 t}$$

① Gjeni koeficientet e seris furie për sinjalet në vazhdim nëse $\omega_0 = 2\pi$

a) $x(t) = 1 + \sin\left(8\pi t + \frac{\pi}{3}\right)$

$$= 1 + \frac{1}{2j} e^{j\left(8\pi t + \frac{\pi}{3}\right)} - \frac{1}{2j} e^{-j\left(8\pi t + \frac{\pi}{3}\right)}$$

$$= \underbrace{1}_{n=0} + \frac{1}{2j} \underbrace{e^{j8\pi t}}_{n=4} \cdot e^{j\frac{\pi}{3}} - \frac{1}{2j} \underbrace{e^{-j8\pi t}}_{n=-4} \cdot e^{-j\frac{\pi}{3}}$$

$$C_0 = 1, \quad C_4 = \frac{1}{2j} e^{j\frac{\pi}{3}}, \quad C_{-4} = -\frac{1}{2j} e^{-j\frac{\pi}{3}}$$

b) $x(t) = [1 + \cos(2\pi t)]\left[\sin\left(10\pi t + \frac{\pi}{6}\right)\right]$

$$= \sin\left(10\pi + \frac{\pi}{6}\right) + \cos(2\pi t)\sin\left(10\pi t + \frac{\pi}{6}\right)$$

$$= \left(\frac{e^{j\pi/6}}{2j} \cdot e^{j2\pi t 5} - \frac{e^{-j\pi/6}}{2j} e^{-j2\pi t 5}\right) + \left(\frac{1}{2} e^{j2\pi t} + \frac{1}{2} e^{-j2\pi t}\right)$$

$$\cdot \left(\frac{e^{j\pi/6}}{2j} e^{j2\pi t 5} - \frac{e^{-j\pi/6}}{2j} e^{-j2\pi t 5}\right)$$

$$= \frac{e^{j\pi/6}}{2j} e^{j2\pi t5} - \frac{e^{-j\pi/6}}{2j} e^{-2\pi t5} + \frac{e^{j\pi/6}}{4j} e^{j2\pi t6} - \frac{e^{-j\pi/6}}{4j} e^{-2\pi t6}$$

$$+ \frac{e^{j\pi/6}}{4j} e^{j2\pi t4} - \frac{e^{-j\pi/6}}{4j} e^{-2\pi t6}.$$

$$C_5 = \frac{e^{j\pi/6}}{2j} \; ; \quad C_{-5} = -\frac{e^{-j\pi/6}}{2j}$$

$$C_6 = \frac{e^{j\pi/6}}{4j} \; ; \quad C_{-4} = -\frac{e^{-j\pi/6}}{4j}$$

$$C_4 = \frac{e^{j\pi/6}}{4j} \; ; \quad C_{-6} = -\frac{e^{j\pi/6}}{4j}$$

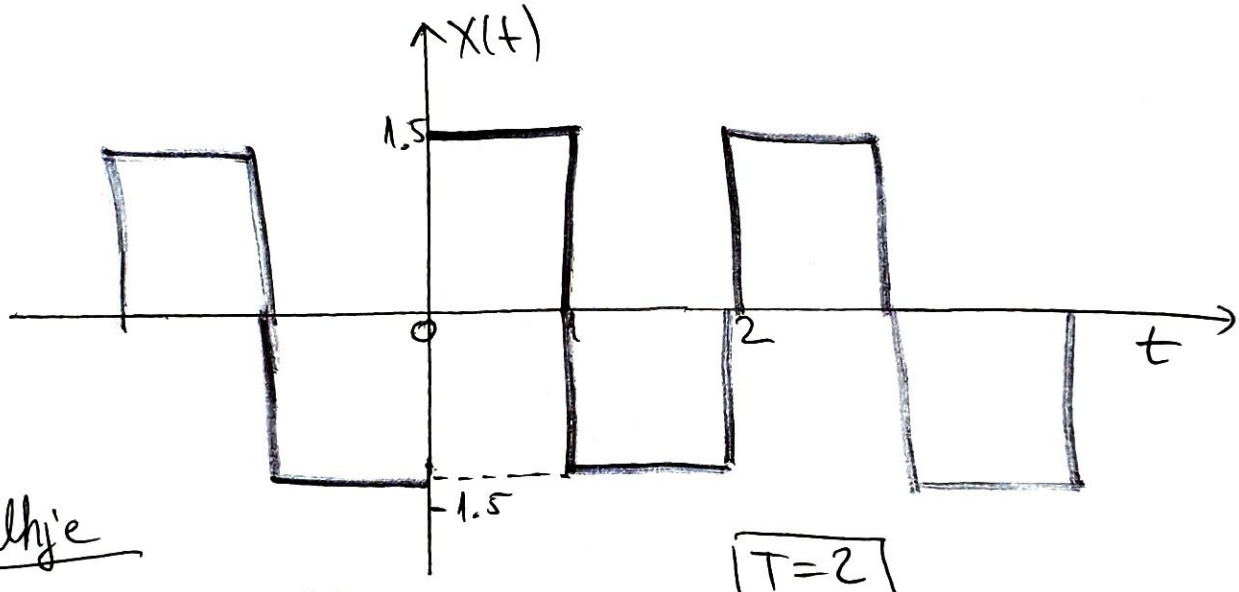
D.P

1. Për sinjalin periodik $X(t)$, vlerat e sinjalit brenda një periode janë dhënë si në vijim:

$$X(t) = \begin{cases} 1,5 & 0 < t < 1 \\ -1,5 & 1 \leq t < 2 \end{cases}$$

Përcaktoni koeficientët e Serisë eksponenciale komplekse jurie C_n.

Zgjidhje



Zgjidhje

$T = 2$

$$\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{2} = \pi$$

$$C_n = \frac{1}{2}\left[\int_{-1}^{0} \left(-\frac{3}{2}\right) e^{-jn\omega_0 t}\, dt + \int_{0}^{1} \frac{3}{2} e^{-jn\omega_0 t}\, dt\right]$$

$$= -\frac{3}{4}\left[-\frac{1}{jn\omega_0}\left(1 - e^{jn\omega_0}\right)\right] + \frac{3}{4}\left[-\frac{1}{jn\omega_0}\left(e^{-jn\omega_0} - 1\right)\right]$$

$$= \frac{3}{4jn\omega_0}\left(1 - e^{jn\omega_0}\right) - \frac{3}{4jn\omega_0}\left(e^{-jn\omega_0} - 1\right)$$

$\boxed{\omega_0 = \pi}$

$$= \frac{3}{4jn\pi}\left(1 - e^{jn\pi}\right) - \frac{3}{4jn\pi}\left(e^{-jn\pi} - 1\right)$$

$$= \left(\frac{3}{4jn\pi}\right) - \frac{3}{4jn\pi}e^{jn\pi} - \frac{3}{4jn\pi}e^{-jn\pi} + \left(\frac{3}{4jn\pi}\right)$$

mbledhim

$$= \frac{3}{2jn\pi} - \frac{3}{2jn\pi}\left(\frac{e^{jn\pi} + e^{-jn\pi}}{2}\right) \rightarrow \cos n\pi \rightarrow (-1)^n$$

$$= \frac{3}{2jn\pi} - \frac{3}{2jn\pi}(-1)^n = \boxed{\frac{3}{2jn\pi}\left(1 - (-1)^n\right).}$$

$\text{Re}\{C_n\} = 0$

$$|C_n| = \frac{3}{2n\pi}\left[1 - (-1)^n\right]$$

kompleksre = Imagjinare